Chapter Contents

Chapter 15 - Matrices

The preceding chapters studied one unknown function at a time. Later chapters will place several unknown functions into a single vector and write their differential equations as one matrix equation.

A matrix is more than a rectangular storage box. Its dimensions determine which operations are legal, and its multiplication rule describes how several quantities interact.

The main questions are:

- how are rows, columns, entries, and vectors described precisely?
- when can matrices be added or multiplied?
- why is matrix multiplication based on rows and columns?
- why can changing the order of a product change the answer?
- how are powers and the identity matrix defined?
- how are matrices containing functions differentiated and integrated?
- how does a determinant produce a characteristic polynomial?
- what do eigenvalues and eigenvectors say about a matrix?
- how does the Cayley-Hamilton theorem reduce high matrix powers?

All examples, equations, and exercises in this chapter are independently constructed. The reference text is used only to identify the mathematical topics.

Reading Matrix Notation

The Goal Of This Block

The goal is to read a matrix unambiguously. Before performing an operation, the reader should be able to state the matrix's size and locate any requested entry.

Rows, Columns, And Dimensions

A matrix is a rectangular arrangement of entries. For example:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

Matrix A has:

- two horizontal rows
- three vertical columns

Its dimensions are therefore:

$$\boxed{2 \times 3}.$$

The order matters. A 2×3 matrix has two rows and three columns, not three rows and two columns.

The practical meaning:

> **dimensions are the first compatibility check for every matrix operation.**

Entry Notation

For a matrix A, the symbol a_{ij} denotes the entry in:

$$\text{row } i, \qquad \text{column } j.$$

Return to:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

The entry in row 1, column 3 is:

$$\boxed{a_{13} = 4}.$$

The entry in row 2, column 1 is:

$$\boxed{a_{21} = 0}.$$

The first index always names the row. The second index always names the column.

The General Form

An $m \times n$ matrix has m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Here:

- m counts rows
- n counts columns
- a_{ij} is the entry in row i, column j

If $m = n$, the matrix is **square**. A square matrix has the same number of rows and columns.

When Two Matrices Are Equal

Two matrices are equal only when both conditions hold:

1. they have the same dimensions;
2. every pair of corresponding entries is equal.

For example:

$$\begin{bmatrix} x+1 & 4 \\ -2 & y \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ -2 & 3 \end{bmatrix}.$$

Equality of the top-left entries gives:

$$x + 1 = 6.$$

Subtract 1 from both sides:

$$x = 5.$$

Equality of the bottom-right entries gives:

$$y = 3.$$

Therefore the matrices are equal when:

$$\boxed{x = 5, \qquad y = 3}.$$

Column Vectors And Row Vectors

A matrix with one column is a **column vector**:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Its dimensions are 3×1.

A matrix with one row is a **row vector**:

$$\mathbf{r} = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix}.$$

Its dimensions are 1×3.

These contain the same number of entries, but they do not have the same shape:

$$3 \times 1 \neq 1 \times 3.$$

In systems of differential equations, unknowns are usually collected into a column vector such as:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

Zero, Diagonal, And Identity Matrices

A **zero matrix** has zero in every position. Its dimensions must still be specified. For example:

$$0_{2\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

A **diagonal matrix** is square and has zero away from its main diagonal:

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

The **identity matrix** has 1 on the main diagonal and 0 elsewhere:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Its role in multiplication is similar to the role of the number 1 in ordinary arithmetic. That property will be verified after matrix multiplication has been defined.

The Transpose

The transpose of A, written A^{T}, turns rows into columns.

Start with:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

The first row of A becomes the first column of A^{T}, and the second row becomes the second column:

$$A^{\mathsf{T}} = \begin{bmatrix} 2 & 0 \\ -1 & 3 \\ 4 & 5 \end{bmatrix}.$$

Thus a 2×3 matrix becomes a 3×2 matrix.

Block 1 Summary

For an $m \times n$ matrix:

$$a_{ij} = \text{entry in row } i \text{ and column } j.$$

Dimensions are written as rows by columns. Vectors are one-row or one-column matrices, square matrices have equal row and column counts, and transposition interchanges rows with columns.

Addition And Scalar Multiplication

The Goal Of This Block

The goal is to combine matrices entry by entry and to distinguish this process from matrix multiplication.

The Size Rule For Addition

Matrices can be added only when they have the same dimensions.

If:

$$A = [a_{ij}], \qquad B = [b_{ij}]$$

are both $m \times n$, then:

$$\boxed{A + B = [a_{ij} + b_{ij}]}.$$

Each entry is added only to the entry occupying the same position.

A 2×3 matrix cannot be added to a 3×2 matrix. Having the same number of entries is not enough; the shapes must match.

A Complete Addition Example

Let:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} -3 & 6 & 1 \\ 7 & -2 & 0 \end{bmatrix}.$$

Both matrices are 2×3, so $A + B$ is defined and will also be 2×3.

Add corresponding entries:

$$\begin{aligned} A + B &= \begin{bmatrix} 2 + (-3) & -1 + 6 & 4 + 1 \\ 0 + 7 & 3 + (-2) & 5 + 0 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 5 & 5 \\ 7 & 1 & 5 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{A + B = \begin{bmatrix} -1 & 5 & 5 \\ 7 & 1 & 5 \end{bmatrix}}.$$

Scalar Multiplication

A scalar is a single number or scalar-valued function. Multiplying a matrix by a scalar multiplies every entry.

Using:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix},$$

multiply every entry by 3:

$$3A = \begin{bmatrix} 3(2) & 3(-1) & 3(4) \\ 3(0) & 3(3) & 3(5) \end{bmatrix}$$
$$= \begin{bmatrix} 6 & -3 & 12 \\ 0 & 9 & 15 \end{bmatrix}.$$

Scalar multiplication does not change the matrix dimensions.

A Matrix Linear Combination

Continue with:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} -3 & 6 & 1 \\ 7 & -2 & 0 \end{bmatrix}.$$

Find $3A - 2B$.

First calculate $3A$:

$$3A = \begin{bmatrix} 6 & -3 & 12 \\ 0 & 9 & 15 \end{bmatrix}.$$

Now calculate $2B$:

$$2B = \begin{bmatrix} -6 & 12 & 2 \\ 14 & -4 & 0 \end{bmatrix}.$$

Subtract corresponding entries:

$$3A - 2B = \begin{bmatrix} 6-(-6) & -3-12 & 12-2 \\ 0-14 & 9-(-4) & 15-0 \end{bmatrix}$$
$$= \begin{bmatrix} 12 & -15 & 10 \\ -14 & 13 & 15 \end{bmatrix}.$$

The completed linear combination is:

$$\boxed{3A - 2B = \begin{bmatrix} 12 & -15 & 10 \\ -14 & 13 & 15 \end{bmatrix}}.$$

Additive Properties

For matrices of compatible dimensions:

$$A + B = B + A$$

and:

$$(A + B) + C = A + (B + C).$$

The zero matrix is the additive identity:

$$A + 0 = A.$$

The matrix $-A$ is formed by negating every entry, so:

$$A + (-A) = 0.$$

These properties work because matrix addition is ordinary addition performed independently in each position.

A Useful Vector Interpretation

Let:

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}.$$

Both are 3×1, so $2\mathbf{u} - \mathbf{v}$ is defined.

First:

$$2\mathbf{u} = \begin{bmatrix} 2 \\ -4 \\ 8 \end{bmatrix}.$$

Then:

$$\begin{aligned} 2\mathbf{u} - \mathbf{v} &= \begin{bmatrix} 2-3 \\ -4-5 \\ 8-(-1) \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -9 \\ 9 \end{bmatrix}. \end{aligned}$$

What this tells us:

vector linear combinations use the same entrywise addition and scalar multiplication rules as all other matrices.

Block 2 Summary

Addition requires equal dimensions:

$$A + B = [a_{ij} + b_{ij}].$$

Scalar multiplication affects every entry:

$$\alpha A = [\alpha a_{ij}].$$

Both operations preserve the dimensions of the matrices involved.

Matrix Multiplication

The Goal Of This Block

The goal is to make row-by-column multiplication predictable rather than mysterious. The dimension test will be applied before any arithmetic begins.

The Inner-Dimension Test

Suppose:

$$A \text{ is } m \times n$$

and:

$$B \text{ is } n \times p.$$

Then AB is defined because the two inner dimensions match:

$$(m \times \boxed{n})(\boxed{n} \times p).$$

The product has the two outer dimensions:

$$\boxed{AB \text{ is } m \times p}.$$

If the inner dimensions do not match, the product is not defined.

Why Rows Meet Columns

Let $A = [a_{ij}]$ be $m \times n$ and let $B = [b_{ij}]$ be $n \times p$. The entry in row i, column j of AB is:

$$\boxed{(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}}.$$

This means:

1. take row i from A;
2. take column j from B;
3. multiply corresponding entries;
4. add those products.

The matching inner dimension ensures that the selected row and column contain the same number of entries.

A Complete Product Example

Let:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 3 & 2 \end{bmatrix}.$$

The dimensions are:

$$A : 2 \times 3, \qquad B : 3 \times 2.$$

The inner dimensions match, so:

$$AB \text{ is } 2 \times 2.$$

Find the row 1, column 1 entry:

$$\begin{aligned} (AB)_{11} &= \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\ &= 1(2) + 2(-1) + (-1)(3) \\ &= 2 - 2 - 3 \\ &= -3. \end{aligned}$$

Find the row 1, column 2 entry:

$$\begin{aligned} (AB)_{12} &= \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} \\ &= 1(1) + 2(5) + (-1)(2) \\ &= 1 + 10 - 2 \\ &= 9. \end{aligned}$$

Find the row 2, column 1 entry:

$$
\begin{aligned}
(AB)_{21} &= \begin{bmatrix} 3 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\
&= 3(2) + 0(-1) + 4(3) \\
&= 6 + 0 + 12 \\
&= 18.
\end{aligned}
$$

Find the row 2, column 2 entry:

$$
\begin{aligned}
(AB)_{22} &= \begin{bmatrix} 3 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} \\
&= 3(1) + 0(5) + 4(2) \\
&= 3 + 0 + 8 \\
&= 11.
\end{aligned}
$$

Place each result in its matching position:

$$
\boxed{AB = \begin{bmatrix} -3 & 9 \\ 18 & 11 \end{bmatrix}}.
$$

Changing The Order Changes The Problem

For the same matrices:

$$
A : 2 \times 3, \qquad B : 3 \times 2,
$$

the reverse product has dimensions:

$$(3 \times 2)(2 \times 3).$$

Therefore BA is defined, but it is 3×3, not 2×2.

Compute it row by row:

$$\begin{aligned} BA &= \begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 2(1)+1(3) & 2(2)+1(0) & 2(-1)+1(4) \\ -1(1)+5(3) & -1(2)+5(0) & -1(-1)+5(4) \\ 3(1)+2(3) & 3(2)+2(0) & 3(-1)+2(4) \end{bmatrix} \\ &= \begin{bmatrix} 5 & 4 & 2 \\ 14 & -2 & 21 \\ 9 & 6 & 5 \end{bmatrix}. \end{aligned}$$

Here AB and BA do not even have the same dimensions. In general:

$$\boxed{AB \neq BA}.$$

Matrix multiplication is not commutative.

Matrix Times Vector

Let:

$$M = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

The dimensions are:

$$(2 \times 2)(2 \times 1),$$

so $M\mathbf{x}$ is a 2×1 vector.

Use one row of M at a time:

$$\begin{aligned} M\mathbf{x} &= \begin{bmatrix} 2(4) + (-1)(-2) \\ 1(4) + 3(-2) \end{bmatrix} \\ &= \begin{bmatrix} 8 + 2 \\ 4 - 6 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ -2 \end{bmatrix}. \end{aligned}$$

Thus:

$$\boxed{M\mathbf{x} = \begin{bmatrix} 10 \\ -2 \end{bmatrix}}.$$

The Column-Combination Interpretation

The same product can be understood using the columns of M.

Write:

$$M = \begin{bmatrix} | & | \\ \mathbf{m}_1 & \mathbf{m}_2 \\ | & | \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Then:

$$\boxed{M\mathbf{x} = x_1\mathbf{m}_1 + x_2\mathbf{m}_2}.$$

For the previous example:

$$\mathbf{m}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{m}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \qquad x_1 = 4, \qquad x_2 = -2.$$

Substitute:

$$\begin{aligned} M\mathbf{x} &= 4\begin{bmatrix} 2 \\ 1 \end{bmatrix} - 2\begin{bmatrix} -1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -6 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ -2 \end{bmatrix}. \end{aligned}$$

Another way to see it:

multiplying by a matrix forms a linear combination of that matrix's columns, using the input vector entries as coefficients.

Associativity And Distributivity

Whenever the dimensions make all products meaningful:

$$A(BC) = (AB)C.$$

Thus matrix multiplication is associative.

It also distributes over addition:

$$A(B + C) = AB + AC$$

and:

$$(A + B)C = AC + BC.$$

The order of factors has not been changed in any of these rules.

Block 3 Summary

The product:

$$(m \times n)(n \times p)$$

is defined and has dimensions:

$$m \times p.$$

Each product entry is a row-column dot product. Matrix multiplication is associative and distributive, but it is not generally commutative.

Identity Matrices, Powers, And Algebraic Caution

The Goal Of This Block

The goal is to define powers of square matrices and identify which familiar algebra rules survive matrix multiplication.

The Identity Matrix In A Product

For an $m \times n$ matrix A:

$$I_m A = A$$

and:

$$AI_n = A.$$

The identity on the left must have m rows and columns. The identity on the right must have n rows and columns.

For a square 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Multiply A by I_2:

$$\begin{aligned} AI_2 &= \begin{bmatrix} a(1) + b(0) & a(0) + b(1) \\ c(1) + d(0) & c(0) + d(1) \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ &= A. \end{aligned}$$

Multiplication by the identity leaves the matrix unchanged.

Powers Of A Square Matrix

If A is square and n is a positive integer, define:

$$\boxed{A^n = \underbrace{AA\cdots A}_{n\text{ factors}}}.$$

In particular:

$$A^2 = AA, \qquad A^3 = AAA.$$

The zeroth power is defined by:

$$\boxed{A^0 = I}.$$

A nonsquare matrix does not have an ordinary square A^2, because the inner dimensions of AA would not match.

A Complete Power Example

Let:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}.$$

First calculate $A^2 = AA$:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1)+2(0) & 1(2)+2(3) \\ 0(1)+3(0) & 0(2)+3(3) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}. \end{aligned}$$

Now use $A^3 = A^2 A$:

$$
\begin{aligned}
A^3 &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \\
&= \begin{bmatrix} 1(1) + 8(0) & 1(2) + 8(3) \\ 0(1) + 9(0) & 0(2) + 9(3) \end{bmatrix} \\
&= \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}.
\end{aligned}
$$

The resulting third power is:

$$\boxed{A^3 = \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}}.$$

Matrix Squaring Is Not Entrywise Squaring

For:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix},$$

the actual matrix square is:

$$A^2 = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}.$$

Squaring each entry would instead give:

$$\begin{bmatrix} 1^2 & 2^2 \\ 0^2 & 3^2 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 9 \end{bmatrix},$$

which is different.

What to remember:

the exponent in A^2 means the matrix product AA, not entrywise exponentiation.

Expanding A Matrix Square

Start from:

$$(A+B)^2 = (A+B)(A+B).$$

Distribute without changing the order of any factors:

$$\begin{aligned}(A+B)^2 &= A(A+B) + B(A+B)\\ &= A^2 + AB + BA + B^2.\end{aligned}$$

The correct matrix expansion is:

$$\boxed{(A+B)^2 = A^2 + AB + BA + B^2}.$$

For ordinary numbers, $AB = BA$, so the middle terms combine to $2AB$. For matrices, that combination is valid only when A and B commute.

Nonzero Matrices Can Multiply To Zero

Let:

$$P = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \qquad Q = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}.$$

Neither matrix is the zero matrix. Calculate PQ:

$$PQ = \begin{bmatrix} 1(2) + (-1)(2) & 1(0) + (-1)(0) \\ 1(2) + (-1)(2) & 1(0) + (-1)(0) \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This calculation shows that two nonzero matrices can have a zero product:

$$\boxed{P \neq 0, \quad Q \neq 0, \quad \text{but} \quad PQ = 0}.$$

This is one reason scalar cancellation habits cannot be imported blindly into matrix algebra.

Block 4 Summary

For square matrices:

$$A^0 = I, \qquad A^n = \underbrace{AA \cdots A}_{n \text{ factors}}.$$

Matrix powers use matrix multiplication. Factor order matters, $(A + B)^2$ contains both AB and BA, and a product of nonzero matrices can equal the zero matrix.

Matrices Whose Entries Are Functions

The Goal Of This Block

The goal is to differentiate and integrate matrices entry by entry while preserving the order of factors in product rules.

A Matrix-Valued Function

A matrix may depend on the independent variable t. For example:

$$A(t) = \begin{bmatrix} t^2 & e^t \\ \sin t & 3t \end{bmatrix}.$$

Each position contains a scalar-valued function of t.

The matrix has fixed dimensions 2×2, even though its entries change with time.

Differentiate Entry By Entry

For:

$$A(t) = [a_{ij}(t)],$$

define:

$$\boxed{A'(t) = [a'_{ij}(t)]}.$$

Using:

$$A(t) = \begin{bmatrix} t^2 & e^t \\ \sin t & 3t \end{bmatrix},$$

differentiate each entry in its current position:

$$A'(t) = \begin{bmatrix} \dfrac{d}{dt}(t^2) & \dfrac{d}{dt}(e^t) \\ \dfrac{d}{dt}(\sin t) & \dfrac{d}{dt}(3t) \end{bmatrix} = \begin{bmatrix} 2t & e^t \\ \cos t & 3 \end{bmatrix}.$$

Thus:

$$\boxed{A'(t) = \begin{bmatrix} 2t & e^t \\ \cos t & 3 \end{bmatrix}}.$$

Integrate Entry By Entry

An indefinite matrix integral is also calculated entry by entry:

$$\int A(t)\,dt = \left[\int a_{ij}(t)\,dt\right].$$

For the same $A(t)$:

$$\int A(t)\,dt = \begin{bmatrix} \int t^2\,dt & \int e^t\,dt \\ \int \sin t\,dt & \int 3t\,dt \end{bmatrix} = \begin{bmatrix} \dfrac{t^3}{3} & e^t \\ -\cos t & \dfrac{3t^2}{2} \end{bmatrix} + C.$$

Here C is a constant matrix of the same dimensions:

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Each entry may require its own integration constant. Writing one scalar constant would incorrectly force all four constants to be equal.

The Matrix Product Rule

If $A(t)B(t)$ is defined for every t, then:

$$\boxed{\frac{d}{dt}[A(t)B(t)] = A'(t)B(t) + A(t)B'(t)}.$$

The order is essential. The first term is $A'B$, and the second is AB'. Neither factor is moved past the other.

A Complete Product-Rule Check

Let:

$$A(t) = \begin{bmatrix} t & 1 \\ 0 & t^2 \end{bmatrix}, \qquad B(t) = \begin{bmatrix} 1 & t \\ t & 0 \end{bmatrix}.$$

First multiply the matrices:

$$\begin{aligned} A(t)B(t) &= \begin{bmatrix} t(1) + 1(t) & t(t) + 1(0) \\ 0(1) + t^2(t) & 0(t) + t^2(0) \end{bmatrix} \\ &= \begin{bmatrix} 2t & t^2 \\ t^3 & 0 \end{bmatrix}. \end{aligned}$$

Differentiate this product entry by entry:

$$\frac{d}{dt}[AB] = \begin{bmatrix} 2 & 2t \\ 3t^2 & 0 \end{bmatrix}.$$

Now verify the product rule directly. Differentiate each factor:

$$A'(t) = \begin{bmatrix} 1 & 0 \\ 0 & 2t \end{bmatrix}, \qquad B'(t) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Calculate the first product-rule term:

$$\begin{aligned} A'B &= \begin{bmatrix} 1 & 0 \\ 0 & 2t \end{bmatrix} \begin{bmatrix} 1 & t \\ t & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ 2t^2 & 0 \end{bmatrix}. \end{aligned}$$

Calculate the second product-rule term:

$$\begin{aligned} AB' &= \begin{bmatrix} t & 1 \\ 0 & t^2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ t^2 & 0 \end{bmatrix}. \end{aligned}$$

Add corresponding entries:

$$\begin{aligned} A'B + AB' &= \begin{bmatrix} 1 & t \\ 2t^2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & t \\ t^2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2t \\ 3t^2 & 0 \end{bmatrix}. \end{aligned}$$

This agrees with the derivative of the product calculated first.

A Matrix Initial-Value Calculation

Suppose:

$$X'(t) = \begin{bmatrix} 2t & 1 \\ \cos t & -e^{-t} \end{bmatrix}$$

and:

$$X(0) = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}.$$

Integrate from 0 to t:

$$X(t) - X(0) = \int_0^t X'(s)\,ds.$$

Calculate the integral entry by entry:

$$\begin{aligned} \int_0^t X'(s)\,ds &= \begin{bmatrix} \int_0^t 2s\,ds & \int_0^t 1\,ds \\ \int_0^t \cos s\,ds & \int_0^t -e^{-s}\,ds \end{bmatrix} \\ &= \begin{bmatrix} t^2 & t \\ \sin t & e^{-t} - 1 \end{bmatrix}. \end{aligned}$$

Add $X(0)$ to both sides:

$$\begin{aligned} X(t) &= \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} t^2 & t \\ \sin t & e^{-t} - 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 + t^2 & -2 + t \\ \sin t & 2 + e^{-t} \end{bmatrix}. \end{aligned}$$

The matrix satisfying the differential equation and initial value is:

$$\boxed{X(t) = \begin{bmatrix} 1+t^2 & -2+t \\ \sin t & 2+e^{-t} \end{bmatrix}}.$$

Check the initial value by setting $t = 0$:

$$X(0) = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}.$$

Block 5 Summary

Matrix differentiation and integration are entrywise operations:

$$A' = [a'_{ij}], \qquad \int A\,dt = \left[\int a_{ij}\,dt\right].$$

The matrix product rule is:

$$(AB)' = A'B + AB',$$

with factor order preserved. An indefinite matrix integral requires a constant matrix, not merely one scalar constant.

Determinants, Eigenvalues, And Eigenvectors

The Goal Of This Block

The goal is to build the characteristic polynomial from a determinant and then interpret its roots as special scaling factors of the matrix.

The Two-By-Two Determinant

A determinant is a scalar associated with a square matrix.

For:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is:

$$\boxed{\det(M) = ad - bc}.$$

The products use opposite corners:

$$\text{main-diagonal product} - \text{other-diagonal product}.$$

For example, let:

$$M = \begin{bmatrix} 3 & -2 \\ 5 & 1 \end{bmatrix}.$$

Substitute the entries into $ad - bc$:

$$\begin{aligned} \det(M) &= 3(1) - (-2)(5) \\ &= 3 - (-10) \\ &= 13. \end{aligned}$$

For this matrix, the determinant is:

$$\boxed{\det(M) = 13}.$$

A Three-By-Three Determinant

For:

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

expansion along the first row gives:

$$\boxed{\det(M) = a(ei - fh) - b(di - fg) + c(dh - eg)}.$$

The cofactor signs across the first row are:

$$+, \quad -, \quad +.$$

Consider:

$$M = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Substitute the first-row entries and their 2×2 determinants:

$$\begin{aligned} \det(M) &= 2[3(1) - 2(0)] - 1[(-1)(1) - 2(4)] \\ &\quad + 0[(-1)(0) - 3(4)]. \end{aligned}$$

Simplify each bracket:

$$\begin{aligned}\det(M) &= 2(3) - 1(-1-8) + 0 \\ &= 6 - (-9) \\ &= 15.\end{aligned}$$

For this matrix, the three-by-three determinant is:

$$\boxed{\det(M) = 15}.$$

The Characteristic Polynomial

Let A be an $n \times n$ matrix, and let I be the $n \times n$ identity matrix.

The characteristic polynomial used in this chapter is:

$$\boxed{p_A(\lambda) = \det(\lambda I - A)}.$$

Here λ is a scalar variable. The matrix $\lambda I - A$ is formed first, and its determinant then produces an ordinary polynomial in λ.

An **eigenvalue** of A is a root of that polynomial:

$$p_A(\lambda) = 0.$$

Some books define the characteristic polynomial as $\det(A - \lambda I)$. That convention differs by a factor $(-1)^n$ but gives the same eigenvalues. Mixing the two conventions halfway through a calculation causes unnecessary sign errors.

A Complete Two-By-Two Eigenvalue Example

Let:

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

Start with the identity matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Multiply it by λ:

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}.$$

Subtract A entry by entry:

$$\lambda I - A = \begin{bmatrix} \lambda - 4 & -2 \\ -1 & \lambda - 3 \end{bmatrix}.$$

Now take the determinant:

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I - A) \\ &= (\lambda - 4)(\lambda - 3) - (-2)(-1). \end{aligned}$$

Multiply the off-diagonal entries:

$$(-2)(-1) = 2.$$

Substituting this off-diagonal product into the determinant gives:

$$p_A(\lambda) = (\lambda - 4)(\lambda - 3) - 2.$$

Expand the brackets:

$$\begin{aligned} p_A(\lambda) &= \lambda^2 - 7\lambda + 12 - 2 \\ &= \lambda^2 - 7\lambda + 10. \end{aligned}$$

Factor the polynomial:

$$p_A(\lambda) = (\lambda - 5)(\lambda - 2).$$

Set the characteristic polynomial equal to zero:

$$(\lambda - 5)(\lambda - 2) = 0.$$

Hence the eigenvalues are:

$$\boxed{\lambda_1 = 5, \qquad \lambda_2 = 2}.$$

What An Eigenvector Adds

A nonzero vector $\mathbf{v}$ is an eigenvector associated with λ when:

$$\boxed{A\mathbf{v} = \lambda\mathbf{v}}.$$

The matrix may change the vector's length or direction sign, but it does not turn the vector away from its line.

To find an eigenvector, rewrite the equation as:

$$(A - \lambda I)\mathbf{v} = 0.$$

The requirement $\mathbf{v} \neq 0$ matters. The zero vector satisfies $A\mathbf{0} = \lambda\mathbf{0}$ for every λ, so it cannot distinguish one eigenvalue from another.

Finding The Eigenvectors In The Example

Continue with:

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

For $\lambda = 5$:

$$A - 5I = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}.$$

Let:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

Then $(A - 5I)\mathbf{v} = 0$ gives:

$$-v_1 + 2v_2 = 0.$$

Add v_1 to both sides:

$$2v_2 = v_1.$$

Choose $v_2 = 1$. Then $v_1 = 2$, so one eigenvector is:

$$\boxed{\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}}.$$

Verify it locally:

$$\begin{aligned} A\mathbf{v}_1 &= \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} +10 \\ 5 \end{bmatrix} \\ &= 5 \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \end{aligned}$$

For $\lambda = 2$:

$$A - 2I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}.$$

The equation $(A - 2I)\mathbf{v} = 0$ reduces to:

$$v_1 + v_2 = 0.$$

Choose $v_1 = 1$. Then $v_2 = -1$, so one eigenvector is:

$$\boxed{\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}}.$$

Verify:

$$\begin{aligned} A\mathbf{v}_2 &= \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -2 \end{bmatrix} \\ &= 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \end{aligned}$$

Any nonzero scalar multiple of an eigenvector is another eigenvector for the same eigenvalue.

Why Eigenvalues Matter For Differential Equations

Consider the future matrix differential equation:

$$\mathbf{x}' = A\mathbf{x}.$$

Suppose $A\mathbf{v} = \lambda\mathbf{v}$. Try:

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}.$$

Differentiate:

$$\mathbf{x}'(t) = \lambda e^{\lambda t}\mathbf{v}.$$

Now multiply $\mathbf{x}$ by A:

$$\begin{aligned} A\mathbf{x}(t) &= A\left(e^{\lambda t}\mathbf{v}\right) \\ &= e^{\lambda t} A\mathbf{v} \\ &= e^{\lambda t}\lambda\mathbf{v} \\ &= \lambda e^{\lambda t}\mathbf{v}. \end{aligned}$$

Both $\mathbf{x}'$ and $A\mathbf{x}$ equal $\lambda e^{\lambda t}\mathbf{v}$. Hence the proposed function satisfies:

$$\mathbf{x}' = A\mathbf{x}.$$

The key intuition:

> **an eigenvector supplies a fixed direction, while its eigenvalue controls the exponential behavior along that direction.**

A later chapter will develop this method fully.

Triangular Matrices And Multiplicity

For a triangular matrix, the characteristic polynomial is the product of λ minus the diagonal entries.

Consider:

$$T = \begin{bmatrix} 3 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & -1 \end{bmatrix}.$$

Then:

$$\lambda I - T = \begin{bmatrix} \lambda - 3 & -1 & 2 \\ 0 & \lambda + 1 & -4 \\ 0 & 0 & \lambda + 1 \end{bmatrix}.$$

This is triangular, so its determinant is the product of its diagonal entries:

$$\begin{aligned} p_T(\lambda) &= (\lambda - 3)(\lambda + 1)(\lambda + 1) \\ &= (\lambda - 3)(\lambda + 1)^2. \end{aligned}$$

The factorization $p_T(\lambda) = (\lambda - 3)(\lambda + 1)^2$ shows that:

$$\boxed{\lambda = 3 \text{ has multiplicity } 1}$$

and:

$$\boxed{\lambda = -1 \text{ has multiplicity } 2}.$$

The exponent in the characteristic polynomial gives the **algebraic multiplicity**. It does not automatically tell us how many independent eigenvectors belong to that eigenvalue.

Block 6 Summary

Using the monic convention:

$$p_A(\lambda) = \det(\lambda I - A).$$

The roots of $p_A(\lambda) = 0$ are the eigenvalues. An associated nonzero vector satisfies:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Eigenpairs matter in ODEs because they generate modes of the form $e^{\lambda t}\mathbf{v}$.

Cayley-Hamilton And A Reliable Matrix Workflow

The Goal Of This Block

The goal is to understand what it means for a matrix to satisfy its own characteristic equation and to use that identity to reduce high powers.

Substituting A Matrix Into A Polynomial

Suppose:

$$p(\lambda) = \lambda^2 - 4\lambda + 7.$$

Replacing the scalar variable λ by a square matrix A gives:

$$p(A) = A^2 - 4A + 7I.$$

The constant term becomes $7I$, not the scalar 7, because every term in the sum must be a matrix of the same dimensions.

More generally, if:

$$p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0,$$

then:

$$p(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1A + c_0I.$$

The Cayley-Hamilton Theorem

Let $p_A(\lambda)$ be the characteristic polynomial of a square matrix A. The Cayley-Hamilton theorem states:

$$\boxed{p_A(A) = 0}.$$

The zero on the right is the zero matrix of the same dimensions as A.

This does not say that $p_A(\lambda) = 0$ for every scalar λ. It says that the matrix expression obtained by replacing λ with A equals the zero matrix.

A Complete Cayley-Hamilton Verification

Let:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}.$$

Construct $\lambda I - A$:

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & -2 \\ -3 & \lambda \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_A(\lambda) &= (\lambda - 1)\lambda - (-2)(-3) \\ &= \lambda^2 - \lambda - 6. \end{aligned}$$

Therefore the characteristic equation is:

$$\lambda^2 - \lambda - 6 = 0.$$

Cayley-Hamilton predicts:

$$A^2 - A - 6I = 0.$$

Calculate A^2 from the matrix product AA:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(3) & 1(2) + 2(0) \\ 3(1) + 0(3) & 3(2) + 0(0) \end{bmatrix} \\ &= \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix}. \end{aligned}$$

Now subtract A:

$$\begin{aligned} A^2 - A &= \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}. \end{aligned}$$

Write $6I$ explicitly:

$$6I = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}.$$

The two displayed matrices are equal, so:

$$A^2 - A = 6I.$$

Subtract $6I$ from both sides:

$$\boxed{A^2 - A - 6I = 0}.$$

This verifies the theorem for the chosen matrix.

Reducing A High Matrix Power

The verified identity is:

$$A^2 = A + 6I.$$

Use it to find A^5 without performing four full matrix multiplications.

Multiply the identity by A:

$$A^3 = A(A + 6I).$$

Distribute:

$$A^3 = A^2 + 6A.$$

Replace A^2 by $A + 6I$:

$$\begin{aligned} A^3 &= (A + 6I) + 6A \\ &= 7A + 6I. \end{aligned}$$

Multiply by A again:

$$A^4 = A(7A + 6I).$$

Distribute:

$$A^4 = 7A^2 + 6A.$$

Replace A^2:

$$\begin{aligned} A^4 &= 7(A + 6I) + 6A \\ &= 7A + 42I + 6A \\ &= 13A + 42I. \end{aligned}$$

Multiply by A once more:

$$A^5 = A(13A + 42I).$$

Distribute:

$$A^5 = 13A^2 + 42A.$$

Replace A^2 by $A + 6I$:

$$\begin{aligned} A^5 &= 13(A + 6I) + 42A \\ &= 13A + 78I + 42A \\ &= 55A + 78I. \end{aligned}$$

Now insert:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}, \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then:

$$\begin{aligned} A^5 &= 55\begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} + 78\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 55 & 110 \\ 165 & 0 \end{bmatrix} + \begin{bmatrix} 78 & 0 \\ 0 & 78 \end{bmatrix} \\ &= \begin{bmatrix} 133 & 110 \\ 165 & 78 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{A^5 = \begin{bmatrix} 133 & 110 \\ 165 & 78 \end{bmatrix}}.$$

The theorem converts every power A^n with $n \geq 2$ into a linear combination of A and I for this 2×2 example.

A Reliable Workflow

Step 1: Write dimensions beside every matrix. Do this before adding or multiplying.

Step 2: Test compatibility. Addition needs identical dimensions. Multiplication needs matching inner dimensions.

Step 3: Predict the output dimensions. For $(m \times n)(n \times p)$, create an empty $m \times p$ result before doing arithmetic.

Step 4: Name the row and column for each product entry. This prevents accidental entrywise multiplication.

Step 5: Preserve factor order. Never commute factors unless commutativity has been proved for those matrices.

Step 6: For eigenvalues, construct $\lambda I - A$ explicitly. Then take the determinant and solve the resulting scalar polynomial.

Step 7: For an eigenvector, solve $(A - \lambda I)\mathbf{v} = 0$. Keep the requirement $\mathbf{v} \neq 0$ visible.

Step 8: For Cayley-Hamilton, convert constants to multiples of I. Then verify or reduce powers one substitution at a time.

Common Mistake: Multiplying Corresponding Entries

Matrix multiplication is not usually entrywise multiplication.

For AB, the entry $(AB)_{ij}$ comes from:

$$\text{row } i \text{ of } A \quad \text{with} \quad \text{column } j \text{ of } B.$$

If corresponding entries are multiplied instead, a different operation has been performed.

Common Mistake: Checking The Outer Dimensions

For:

$$(m \times n)(r \times p),$$

the product exists only when:

$$\boxed{n = r}.$$

The inner dimensions must match. The outer dimensions m and p describe the answer; they do not decide whether multiplication is legal.

Common Mistake: Reversing A Product

From AB, one cannot casually move to BA.

Possible outcomes include:

- both products exist but have different values
- both products exist but have different interpretations
- one product exists while the other does not
- both products exist but have different dimensions

Always retest dimensions after reversing the order.

Common Mistake: Losing Constants In A Matrix Integral

If:

$$X'(t) = A(t),$$

then an indefinite integration gives:

$$X(t) = \int A(t)\,dt + C,$$

where C is a constant matrix. Each entry can have an independent constant.

Common Mistake: Treating An Eigenvalue As A Vector

An eigenvalue λ is a scalar. An eigenvector $\mathbf{v}$ is a nonzero vector. They are linked by:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The equation does not say $A\lambda = \mathbf{v}$ or $A\mathbf{v} = \mathbf{v}$ unless $\lambda = 1$.

Common Mistake: Reading Multiplicity From A Matrix Entry

Algebraic multiplicity comes from the exponent of a factor in the characteristic polynomial.

If:

$$p_A(\lambda) = (\lambda - 4)^3(\lambda + 2),$$

then $\lambda = 4$ has algebraic multiplicity 3. Repeated appearances of the number 4 among the matrix entries would not establish that fact.

Common Mistake: Substituting A Scalar Constant Into Cayley-Hamilton

If:

$$p_A(\lambda) = \lambda^2 - 5\lambda + 6,$$

then:

$$p_A(A) = A^2 - 5A + 6I.$$

The final term is $6I$, because it must be a matrix. Writing $A^2 - 5A + 6$ tries to add a scalar directly to matrices.

Block 7 Summary

The Cayley-Hamilton theorem states:

$$p_A(A) = 0.$$

It turns the characteristic polynomial into a matrix identity. That identity can verify a calculation, reduce high powers, and later simplify functions of matrices. Reliable work begins with dimensions, preserves factor order, and keeps scalar, vector, and matrix objects distinct.

Original Practice Set

Practice Problems

Problem 1: Read Entries And Form A Transpose. Suppose:

$$A = \begin{bmatrix} 2 & a & -1 \\ b & 4 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 & -1 \\ -3 & 4 & 0 \end{bmatrix}.$$

Find a and b, state the dimensions of A, and write A^{T}.

Problem 2: Calculate A Linear Combination. Let:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} -2 & 5 & 1 \\ 3 & -1 & 2 \end{bmatrix}.$$

Find $2A - 3B$.

Problem 3: Test Product Dimensions. Suppose:

$$A : 3 \times 2, \qquad B : 2 \times 4, \qquad C : 4 \times 3.$$

For each expression, state whether it is defined. If it is defined, state its dimensions:

$$AB, \qquad BA, \qquad BC, \qquad CA, \qquad AC, \qquad (AB)C.$$

Problem 4: Calculate A Rectangular Product. Let:

$$P = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}, \qquad Q = \begin{bmatrix} 1 & 2 \\ -2 & 0 \\ 3 & -1 \end{bmatrix}.$$

Find PQ. Show the row-column calculation for every entry.

Problem 5: Compare Both Product Orders. Let:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix}.$$

Find AB and BA. Use the results to decide whether these matrices commute.

Problem 6: Multiply A Matrix By A Vector. Let:

$$M = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}.$$

Find $M\mathbf{x}$ by row-column multiplication. Then reproduce the result as a linear combination of the columns of M.

Problem 7: Calculate Matrix Powers. Let:

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

Find A^2 and A^3. Explain briefly why squaring each entry would not produce A^2.

Problem 8: Differentiate And Integrate A Matrix. Let:

$$A(t) = \begin{bmatrix} t^3 & \cos t \\ e^{-t} & 2t - 1 \end{bmatrix}.$$

Find:

$$A'(t)$$

and:

$$\int_0^1 A(t)\,dt.$$

Problem 9: Use The Matrix Product Rule. Let:

$$A(t) = \begin{bmatrix} t & 1 \\ 0 & t \end{bmatrix}, \qquad B(t) = \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix}.$$

Use:

$$(AB)' = A'B + AB'$$

to find $\frac{d}{dt}[A(t)B(t)]$. Then check the result by multiplying AB first and differentiating afterward.

Problem 10: Find Eigenvalues And Eigenvectors. Let:

$$C = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

Find the characteristic polynomial, both eigenvalues, and one eigenvector for each eigenvalue.

Problem 11: Read Multiplicity From A Triangular Matrix. Let:

$$T = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & -3 \end{bmatrix}.$$

Find the characteristic polynomial and list each eigenvalue with its algebraic multiplicity.

Problem 12: Verify Cayley-Hamilton And Reduce A Power. Let:

$$D = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}.$$

1. Find $p_D(\lambda) = \det(\lambda I - D)$.
2. Verify $p_D(D) = 0$.
3. Use the resulting matrix identity to find D^4.

Worked Answers: Problems 1 To 4

Answer 1. Two matrices are equal when corresponding entries are equal. Compare the row 1, column 2 entries:

$$a = 5.$$

Compare the row 2, column 1 entries:

$$b = -3.$$

The matrix has two rows and three columns, so:

$$\boxed{A \text{ is } 2 \times 3}.$$

Substitute $a = 5$ and $b = -3$:

$$A = \begin{bmatrix} 2 & 5 & -1 \\ -3 & 4 & 0 \end{bmatrix}.$$

To transpose A, turn each row into a column. The first row becomes:

$$\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix},$$

and the second row becomes:

$$\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}.$$

Placing the two row vectors of A into columns gives:

$$\boxed{A^{\mathsf{T}} = \begin{bmatrix} 2 & -3 \\ 5 & 4 \\ -1 & 0 \end{bmatrix}}.$$

The transpose has dimensions 3×2.

Answer 2. The requested expression is:

$$2A - 3B.$$

First multiply every entry of A by 2:

$$\begin{aligned} 2A &= 2\begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -4 & 6 \\ 8 & 0 & -2 \end{bmatrix}. \end{aligned}$$

Now multiply every entry of B by 3:

$$\begin{aligned} 3B &= 3\begin{bmatrix} -2 & 5 & 1 \\ 3 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -6 & 15 & 3 \\ 9 & -3 & 6 \end{bmatrix}. \end{aligned}$$

Subtract corresponding entries:

$$\begin{aligned} 2A - 3B &= \begin{bmatrix} 2-(-6) & -4-15 & 6-3 \\ 8-9 & 0-(-3) & -2-6 \end{bmatrix} \\ &= \begin{bmatrix} 8 & -19 & 3 \\ -1 & 3 & -8 \end{bmatrix}. \end{aligned}$$

The completed calculation of $2A - 3B$ gives:

$$\boxed{2A - 3B = \begin{bmatrix} 8 & -19 & 3 \\ -1 & 3 & -8 \end{bmatrix}}.$$

Answer 3. A product:

$$(m \times n)(r \times p)$$

is defined only when the inner dimensions match:

$$n = r.$$

When it is defined, the answer has dimensions $m \times p$.

For AB:

$$(3 \times 2)(2 \times 4).$$

The inner dimensions are both 2, so:

$$\boxed{AB \text{ is defined and is } 3 \times 4}.$$

For BA:

$$(2 \times 4)(3 \times 2).$$

The inner dimensions are 4 and 3, so:

$$\boxed{BA \text{ is not defined}}.$$

For BC:

$$(2 \times 4)(4 \times 3).$$

The inner dimensions are both 4, so:

$$\boxed{BC \text{ is defined and is } 2 \times 3}.$$

For CA:

$$(4 \times 3)(3 \times 2).$$

The inner dimensions are both 3, so:

$$\boxed{CA \text{ is defined and is } 4 \times 2}.$$

For AC:

$$(3 \times 2)(4 \times 3).$$

The inner dimensions are 2 and 4, so:

$$\boxed{AC \text{ is not defined}}.$$

Finally, AB has dimensions 3×4. Therefore:

$$(AB)C \quad \text{has dimensions} \quad (3 \times 4)(4 \times 3).$$

The inner dimensions match, so:

$$\boxed{(AB)C \text{ is defined and is } 3 \times 3}.$$

Answer 4. The dimensions are:

$$P : 2 \times 3, \qquad Q : 3 \times 2.$$

Therefore PQ is defined and will be 2×2.

For row 1, column 1:

$$\begin{aligned}(PQ)_{11} &= 2(1) + (-1)(-2) + 3(3) \\ &= 2 + 2 + 9 \\ &= 13.\end{aligned}$$

For row 1, column 2:

$$\begin{aligned}(PQ)_{12} &= 2(2) + (-1)(0) + 3(-1) \\ &= 4 + 0 - 3 \\ &= 1.\end{aligned}$$

For row 2, column 1:

$$\begin{aligned}(PQ)_{21} &= 0(1) + 4(-2) + 1(3) \\ &= 0 - 8 + 3 \\ &= -5.\end{aligned}$$

For row 2, column 2:

$$\begin{aligned}(PQ)_{22} &= 0(2) + 4(0) + 1(-1)\\ &= 0 + 0 - 1\\ &= -1.\end{aligned}$$

Place these entries in their corresponding positions:

$$\boxed{PQ = \begin{bmatrix} 13 & 1 \\ -5 & -1 \end{bmatrix}}.$$

Worked Answers: Problems 5 To 8

Answer 5. Both matrices are 2×2, so both AB and BA are defined.

Calculate AB:

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1(3) + 2(4) & 1(0) + 2(2) \\ 0(3) + (-1)(4) & 0(0) + (-1)(2) \end{bmatrix} \\ &= \begin{bmatrix} 11 & 4 \\ -4 & -2 \end{bmatrix}. \end{aligned}$$

Now reverse the order:

$$\begin{aligned} BA &= \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1) + 0(0) & 3(2) + 0(-1) \\ 4(1) + 2(0) & 4(2) + 2(-1) \end{bmatrix} \\ &= \begin{bmatrix} 3 & 6 \\ 4 & 6 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{AB = \begin{bmatrix} 11 & 4 \\ -4 & -2 \end{bmatrix} \neq \begin{bmatrix} 3 & 6 \\ 4 & 6 \end{bmatrix} = BA}.$$

The matrices do not commute.

Answer 6. The dimensions are:

$$M : 2 \times 3, \qquad \mathbf{x} : 3 \times 1.$$

Therefore $M\mathbf{x}$ is defined and will be 2×1.

Use the first row of M for the first output entry:

$$\begin{aligned}(M\mathbf{x})_1 &= 2(1) + 1(-2) + (-1)(4) \\ &= 2 - 2 - 4 \\ &= -4.\end{aligned}$$

Use the second row for the second output entry:

$$\begin{aligned}(M\mathbf{x})_2 &= 0(1) + 3(-2) + 2(4) \\ &= 0 - 6 + 8 \\ &= 2.\end{aligned}$$

The row-column calculation gives:

$$\boxed{M\mathbf{x} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}}.$$

Now write the columns of M as:

$$\mathbf{m}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \qquad \mathbf{m}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \qquad \mathbf{m}_3 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

The entries of $\mathbf{x}$ are 1, −2, and 4. Therefore:

$$M\mathbf{x} = 1\mathbf{m}_1 - 2\mathbf{m}_2 + 4\mathbf{m}_3.$$

Substitute the columns:

$$\begin{aligned} M\mathbf{x} &= 1\begin{bmatrix}2\\0\end{bmatrix} - 2\begin{bmatrix}1\\3\end{bmatrix} + 4\begin{bmatrix}-1\\2\end{bmatrix} \\ &= \begin{bmatrix}2\\0\end{bmatrix} + \begin{bmatrix}-2\\-6\end{bmatrix} + \begin{bmatrix}-4\\8\end{bmatrix} \\ &= \begin{bmatrix}-4\\2\end{bmatrix}. \end{aligned}$$

The column-combination interpretation gives the same result.

Answer 7. A matrix power uses matrix multiplication:

$$A^2 = AA.$$

Substitute:

$$A = \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}.$$

Then:

$$\begin{aligned} A^2 &= \begin{bmatrix}2 & 1\\0 & 2\end{bmatrix}\begin{bmatrix}2 & 1\\0 & 2\end{bmatrix} \\ &= \begin{bmatrix}2(2)+1(0) & 2(1)+1(2)\\0(2)+2(0) & 0(1)+2(2)\end{bmatrix} \\ &= \begin{bmatrix}4 & 4\\0 & 4\end{bmatrix}. \end{aligned}$$

The matrix product AA gives:

$$\boxed{A^2 = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}}.$$

Now use:

$$A^3 = A^2 A.$$

Substitute the calculated A^2 and the original A:

$$\begin{aligned} A^3 &= \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4(2) + 4(0) & 4(1) + 4(2) \\ 0(2) + 4(0) & 0(1) + 4(2) \end{bmatrix} \\ &= \begin{bmatrix} 8 & 12 \\ 0 & 8 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{A^3 = \begin{bmatrix} 8 & 12 \\ 0 & 8 \end{bmatrix}}.$$

Entrywise squaring would produce:

$$\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix},$$

which differs from A^2. The notation A^2 means AA.

Answer 8. Differentiate each entry of:

$$A(t) = \begin{bmatrix} t^3 & \cos t \\ e^{-t} & 2t - 1 \end{bmatrix}.$$

The entrywise derivative is:

$$\begin{aligned} A'(t) &= \begin{bmatrix} \frac{d}{dt}(t^3) & \frac{d}{dt}(\cos t) \\ \frac{d}{dt}(e^{-t}) & \frac{d}{dt}(2t-1) \end{bmatrix} \\ &= \begin{bmatrix} 3t^2 & -\sin t \\ -e^{-t} & 2 \end{bmatrix}. \end{aligned}$$

Thus:

$$\boxed{A'(t) = \begin{bmatrix} 3t^2 & -\sin t \\ -e^{-t} & 2 \end{bmatrix}}.$$

For the definite integral, integrate each entry from 0 to 1:

$$\int_0^1 A(t)\,dt = \begin{bmatrix} \int_0^1 t^3\,dt & \int_0^1 \cos t\,dt \\ \int_0^1 e^{-t}\,dt & \int_0^1 (2t-1)\,dt \end{bmatrix}.$$

Calculate the top-left entry:

$$\begin{aligned} \int_0^1 t^3\,dt &= \left[\frac{t^4}{4}\right]_0^1 \\ &= \frac{1}{4}. \end{aligned}$$

Calculate the top-right entry:

$$\begin{aligned}\int_0^1 \cos t\,dt &= [\sin t]_0^1\\ &= \sin 1.\end{aligned}$$

Calculate the bottom-left entry:

$$\begin{aligned}\int_0^1 e^{-t}\,dt &= [-e^{-t}]_0^1\\ &= -e^{-1} - (-1)\\ &= 1 - e^{-1}.\end{aligned}$$

Calculate the bottom-right entry:

$$\begin{aligned}\int_0^1 (2t-1)\,dt &= [t^2 - t]_0^1\\ &= (1-1) - (0-0)\\ &= 0.\end{aligned}$$

Place the four results in their original positions:

$$\boxed{\int_0^1 A(t)\,dt = \begin{bmatrix} \frac{1}{4} & \sin 1 \\ 1 - e^{-1} & 0 \end{bmatrix}}.$$

A definite integral does not require an arbitrary constant matrix.

Worked Answers: Problems 9 To 12

Answer 9. The product rule is:

$$(AB)' = A'B + AB'.$$

Differentiate $A(t)$ entry by entry:

$$A'(t) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Differentiate $B(t)$ entry by entry:

$$B'(t) = \begin{bmatrix} e^t & 0 \\ 1 & 0 \end{bmatrix}.$$

Calculate the first product-rule term:

$$\begin{aligned} A'B &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix} \\ &= \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix}. \end{aligned}$$

Calculate the second term:

$$\begin{aligned} AB' &= \begin{bmatrix} t & 1 \\ 0 & t \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} te^t + 1 & 0 \\ t & 0 \end{bmatrix}. \end{aligned}$$

Add corresponding entries:

$$\begin{aligned} A'B + AB' &= \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix} + \begin{bmatrix} te^t + 1 & 0 \\ t & 0 \end{bmatrix} \\ &= \begin{bmatrix} (t+1)e^t + 1 & 0 \\ 2t & 1 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{\frac{d}{dt}[A(t)B(t)] = \begin{bmatrix} (t+1)e^t + 1 & 0 \\ 2t & 1 \end{bmatrix}}.$$

To check, multiply first:

$$\begin{aligned} A(t)B(t) &= \begin{bmatrix} t & 1 \\ 0 & t \end{bmatrix} \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix} \\ &= \begin{bmatrix} te^t + t & 1 \\ t^2 & t \end{bmatrix}. \end{aligned}$$

Differentiate the product entry by entry:

$$\frac{d}{dt}[AB] = \begin{bmatrix} (t+1)e^t + 1 & 0 \\ 2t & 1 \end{bmatrix}.$$

The two methods agree.

Answer 10. The characteristic polynomial is:

$$p_C(\lambda) = \det(\lambda I - C).$$

Construct the matrix:

$$\lambda I - C = \begin{bmatrix} \lambda - 5 & -2 \\ -2 & \lambda - 2 \end{bmatrix}.$$

Take its determinant:

$$\begin{aligned} p_C(\lambda) &= (\lambda - 5)(\lambda - 2) - (-2)(-2) \\ &= (\lambda - 5)(\lambda - 2) - 4. \end{aligned}$$

Expand:

$$\begin{aligned} p_C(\lambda) &= \lambda^2 - 7\lambda + 10 - 4 \\ &= \lambda^2 - 7\lambda + 6. \end{aligned}$$

Factor:

$$p_C(\lambda) = (\lambda - 6)(\lambda - 1).$$

Therefore:

$$\boxed{\lambda_1 = 6, \qquad \lambda_2 = 1}.$$

For $\lambda = 6$, solve:

$$(C - 6I)\mathbf{v} = 0.$$

The coefficient matrix is:

$$C - 6I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}.$$

The first row gives:

$$-v_1 + 2v_2 = 0.$$

Hence:

$$v_1 = 2v_2.$$

Choose $v_2 = 1$, which gives $v_1 = 2$. One eigenvector is:

$$\boxed{\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}}.$$

Check:

$$\begin{aligned} C\mathbf{v}_1 &= \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 12 \\ 6 \end{bmatrix} \\ &= 6 \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \end{aligned}$$

For $\lambda = 1$, solve:

$$(C - I)\mathbf{v} = 0.$$

The coefficient matrix is:

$$C - I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$

The second row gives:

$$2v_1 + v_2 = 0.$$

Therefore:

$$v_2 = -2v_1.$$

Choose $v_1 = 1$, so one eigenvector is:

$$\boxed{\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}}.$$

Check:

$$\begin{aligned} C\mathbf{v}_2 &= \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= 1 \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \end{aligned}$$

Answer 11. The characteristic polynomial is:

$$p_T(\lambda) = \det(\lambda I - T).$$

Construct:

$$\lambda I - T = \begin{bmatrix} \lambda - 4 & -2 & 0 \\ 0 & \lambda - 4 & 1 \\ 0 & 0 & \lambda + 3 \end{bmatrix}.$$

This matrix is triangular. Its determinant is the product of its diagonal entries:

$$\begin{aligned} p_T(\lambda) &= (\lambda - 4)(\lambda - 4)(\lambda + 3) \\ &= (\lambda - 4)^2(\lambda + 3). \end{aligned}$$

Therefore:

$$\boxed{\lambda = 4 \text{ has algebraic multiplicity } 2}$$

and:

$$\boxed{\lambda = -3 \text{ has algebraic multiplicity } 1}.$$

The off-diagonal entries do not change the fact that a triangular matrix's eigenvalues are its diagonal entries.

Answer 12. The characteristic polynomial is:

$$p_D(\lambda) = \det(\lambda I - D).$$

For:

$$D = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix},$$

construct:

$$\lambda I - D = \begin{bmatrix} \lambda & -2 \\ -1 & \lambda - 3 \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_D(\lambda) &= \lambda(\lambda-3)-(-2)(-1) \\ &= \lambda^2-3\lambda-2. \end{aligned}$$

Thus:

$$\boxed{p_D(\lambda) = \lambda^2-3\lambda-2}.$$

Cayley-Hamilton predicts:

$$D^2-3D-2I=0.$$

Calculate D^2:

$$\begin{aligned} D^2 &= \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 0(0)+2(1) & 0(2)+2(3) \\ 1(0)+3(1) & 1(2)+3(3) \end{bmatrix} \\ &= \begin{bmatrix} 2 & 6 \\ 3 & 11 \end{bmatrix}. \end{aligned}$$

Calculate $3D$:

$$3D = \begin{bmatrix} 0 & 6 \\ 3 & 9 \end{bmatrix}.$$

Subtract:

$$\begin{aligned} D^2-3D &= \begin{bmatrix} 2 & 6 \\ 3 & 11 \end{bmatrix} - \begin{bmatrix} 0 & 6 \\ 3 & 9 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= 2I. \end{aligned}$$

Subtract $2I$ from both sides:

$$\boxed{D^2 - 3D - 2I = 0}.$$

Rearrange the verified identity:

$$D^2 = 3D + 2I.$$

Multiply both sides by D:

$$D^3 = 3D^2 + 2D.$$

Replace D^2 with $3D + 2I$:

$$\begin{aligned} D^3 &= 3(3D + 2I) + 2D \\ &= 9D + 6I + 2D \\ &= 11D + 6I. \end{aligned}$$

Multiply by D again:

$$D^4 = 11D^2 + 6D.$$

Replace D^2 once more:

$$\begin{aligned} D^4 &= 11(3D + 2I) + 6D \\ &= 33D + 22I + 6D \\ &= 39D + 22I. \end{aligned}$$

Insert D and I:

$$\begin{aligned} D^4 &= 39\begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} + 22\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 78 \\ 39 & 117 \end{bmatrix} + \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} \\ &= \begin{bmatrix} 22 & 78 \\ 39 & 139 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{D^4 = \begin{bmatrix} 22 & 78 \\ 39 & 139 \end{bmatrix}}.$$

Chapter 15 Final Summary

An $m \times n$ matrix has m rows and n columns, and a_{ij} denotes its row i, column j entry.

The core compatibility rules are:

$$\begin{aligned} A + B &= [a_{ij} + b_{ij}] \quad \text{when the dimensions match,} \\ (m \times n)(n \times p) &\longrightarrow m \times p, \\ (AB)_{ij} &= \sum_{k=1}^{n} a_{ik} b_{kj}. \end{aligned}$$

Matrix multiplication is associative and distributive, but generally $AB \neq BA$. For square matrices and matrix-valued functions:

$$\begin{aligned} A^0 &= I, & A^n &= \underbrace{AA \cdots A}_{n \text{ factors}}, \\ A' &= [a'_{ij}], & \int A\,dt &= \left[\int a_{ij}\,dt\right] + C, \\ (AB)' &= A'B + AB'. \end{aligned}$$

Using the monic characteristic-polynomial convention:

$$p_A(\lambda) = \det(\lambda I - A).$$

Its roots are eigenvalues. An associated nonzero eigenvector satisfies:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The Cayley-Hamilton theorem converts that scalar polynomial into the matrix identity:

$$p_A(A) = 0.$$

This identity can reduce higher matrix powers.

The chapter's central lesson is:

> **begin with dimensions, preserve factor order, and identify whether each object is a scalar, vector, or matrix before applying an algebraic rule.**